ARC-AGI-2 / Qwen3.8-27B-FP8

Deep Notebook + Evidence Bundle Audit Reference

Kaggle 4 x NVIDIA L4 | PIV solver lineage | Reasoning activation | Throughput | P1 side branch

Reference date: 2026-09-17

Status: Canonical audit reference for the next solver-quality stage

Purpose: এই document-টি uploaded source notebooks, executed notebooks, evidence ZIP, manifests, traces, metrics এবং আগের handoff cross-audit করে তৈরি। এর লক্ষ্য হলো একই ভুল পুনরাবৃত্তি বন্ধ করা এবং C1 real solver smoke-এর আগে এক জায়গায় VERIFIED facts, invalid assumptions, নতুন insights এবং exact next-step protocol lock করা।

Evidence labels: VERIFIED = artifact-এ সরাসরি প্রমাণিত; INFERENCE = একাধিক প্রমাণ থেকে audit synthesis; RETIRED = invalidated/obsolete assumption; UNKNOWN = পরের experiment ছাড়া উত্তর নেই।

1. Executive summary

Current project state: Infrastructure, REPL, verifier, reasoning activation এবং serving bottleneck যথেষ্ট পরিমাণে isolated হয়েছে। এখন primary blocker হলো executable program quality: এখনও reproducible non-zero all-train-exact program প্রমাণিত হয়নি।

সব evidence একত্রে দেখলে project এখন "mechanism works, solver quality unproven" অবস্থায়। XHIGH path tool phase-এ যায়নি; LOW এবং BUDGET3072 tool activation দেখিয়েছে; CUDA graph/noAR serving long decode-এ প্রায় 2.67x speedup দিয়েছে। কিন্তু activation probe-এ final program 0 এবং train-exact 0 ছিল। তাই next decisive experiment serving নয়, real solver-quality smoke।

Area	Audit status	Meaning
PIV harness / verifier	VERIFIED	Persistent REPL, retry semantics, fresh all-train exact gate, separate test execution state, terminal coverage hardening repeatedly preflighted.
Reasoning activation	VERIFIED	ACT-LOW 7/8 এবং ACT-BUDGET3072 7/8 tool engagement; MEDIUM 1/8; XHIGH 0/8.
Serving candidate	VERIFIED	C1 graph/noAR long@8 = 129.17 tok/s বনাম frozen C0 = 48.30 tok/s; custom AR meaningful gain দেয়নি.
Solver quality	UNKNOWN / critical	No reproducible non-zero all-train-exact program yet.
Prefix / TP2	UNKNOWN	Aggressive max_num_seqs=64 OOM দিয়ে feature feasibility নির্ধারণ করা যায় না.
P1 SPARSE	HOLD	দুই run-এর aggregate কাছাকাছি, কিন্তু per-game variance এবং runtime-health issues আছে; main ARC-AGI-2 solver line থেকে আলাদা রাখতে হবে.

1.1 Decision locked by this audit

1. C1 real solver SMOKE বানাতে current v1.9d source donor ব্যবহার করতে হবে।
2. Engine-side একমাত্র causal change হবে enforce_eager=True -> False। disable_custom_all_reduce=True এবং enable_prefix_caching=False থাকবে।
3. প্রথম isolation arm হবে ACT-LOW। কারণ tool engagement BUDGET3072-এর সমান হলেও probe-এ LOW-এর tool execution materially cleaner এবং token cost কম ছিল। এটি INFERENCE, solver-quality verdict নয়।
4. Main success chain: final_program -> syntax/execution valid -> all_train_exact -> valid test execution -> official exact।
5. all_train_exact > 0 না আসা পর্যন্ত Proxy/Public-48, pass@K scaling বা Public-120 চালানো যাবে না।

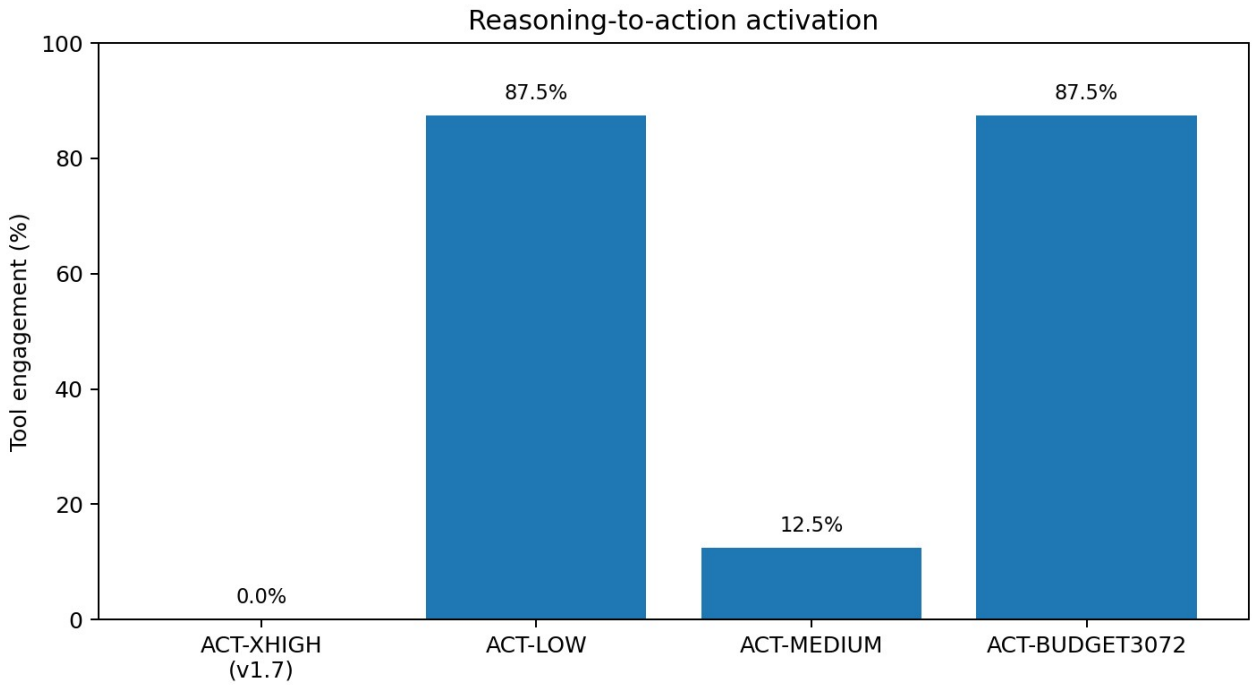


Figure 1. Reasoning-to-action activation. XHIGH থেকে LOW/BUDGET3072-এ mechanism activation-এর বড় পরিবর্তন।

2. Evidence scope and integrity

এই audit-এ current conversation-এর source notebooks, executed notebooks এবং evidence bundles programmatically parse করা হয়েছে। Notebook JSON validity, code-cell compilation, execution/output presence, bundle ZIP integrity, manifests, run IDs, key reports এবং selected raw traces cross-check করা হয়েছে।

Evidence	SHA256 prefix	Integrity	Role
PIV v1.7 SMOKE bundle	7435b795691d...	38/38 manifest files hash-verified	Canonical XHIGH failure reference
ACT v1.9 PREFLIGHT bundle	fab4fcd700d0...	68/68 manifest files hash-verified	Pre-v1.9d preflight evidence
ACT v1.9 PROBE bundle	d400bbb1cbcf...	98/98 manifest files hash-verified	Canonical activation evidence
TP v1.2.4 SMOKE bundle	c34fd3736dd9...	ZIP integrity passed; executed notebook bundle SHA matches handoff	Canonical serving evidence
P1 SPARSE run A	311939cee639...	ZIP integrity and selfcheck hard_complete	Secondary branch
P1 SPARSE run B	d8456de66c3b...	ZIP integrity and selfcheck hard_complete	Secondary branch

Important: CPU SELFTEST bundles are useful for control-flow and packaging tests, but fake-engine throughput values are not hardware evidence. Only real 4xL4 executed runs are used for performance conclusions.

3. PIV solver lineage: what changed and why

Version	Main correction	Why it matters
v1.0	Stateful REPL + fresh exact verifier introduced	Correct core direction, but early implementation still needed source and runtime hardening.
v1.2	Train-perfect state separated from later test execution	Prevents a valid train-perfect result from being erased by a test execution failure.
v1.3	Retry, parser, grid normalization, worker restart, sandbox and finalization hardening	Closed many interface-level failure modes.
v1.4	Inner same-history retry vs outer source-attempt restart corrected	Matched PiVOTools two-level retry semantics.
v1.5	Weak first-turn gate replaced by terminal budget-death + wall-time feasibility	Moved viability gating toward actual terminal behavior.
v1.6	Deadline projection, stratified SMOKE8, guard accounting, sanitization, real scheduler preflight	Removed bias and several budget/accounting bugs.
v1.7	Dynamic multi-call budget fix + unique terminal coverage	Canonical XHIGH SMOKE donor; mechanism clean but model behavior failed before tools.
v1.8 / v1.9	Reasoning-to-action policy ablation	LOW, MEDIUM, BUDGET3072 introduced without changing PIV core.
v1.9d	Record/selftest evidence fixes	stop_without_think_close recorded; fake lineage G actually exercises state reuse; injected-stop token assertion corrected; fake traces bundled.

3.1 A subtle but important preflight lesson

The executed v1.9 PREFLIGHT run 20260915T113233Z reported the fake scheduler selftest as PASS, but its fake lineage G showed stateful_reuse_events=0. In other words, the metric existed but the selftest did not actually exercise it. The later v1.9d source fixes this by defining a symbol in one tool call, reusing it in a later call, asserting the reuse event, and retaining the fake traces. This is a strong example of why "selftest passed" is not sufficient unless the intended property is actively exercised.

VERIFIED source selection: Current PIV-Q27-ACT-v1.9 source notebook SHA 007c8d68559c... এবং executed GPU probe notebook SHA 59cf30009100... এর code-source diff শুধু RUN_PROFILE = PREFLIGHT বনাম PROBE_GPU। অর্থাৎ current source is the correct v1.9d donor for the next derivative.

4. v1.7 XHIGH SMOKE: clean harness, failed model behavior

Run 20260914T122911Z, 8-task stratified SMOKE8, full P0 budget। Infrastructure failure rate ছিল 0 এবং terminal coverage complete ছিল। তবু 8/8 task first turn-এই 12,288 generated-token cap hit করে, tool call 0, final program 0 এবং all-train-exact 0।

Metric	Observed
Tool engagement	0/8 = 0%
Final program	0/8
All-train-exact	0/8
Budget death	8/8 = 100%
First-turn length	8/8 = 100%
Tokens at death	12,288 for every task
Measured sec/output token	0.16014
Projected Proxy48 +20%	4.06 h, but quality gate failed

Finding: This run disproves the idea that parser, REPL or verifier was the dominant blocker. The dominant blocker was the template/budget interaction: XHIGH remained in thinking until the entire task budget was exhausted.

5. v1.9d real GPU activation probe: what was actually proven

Run 20260915T124541Z deliberately used a cheap canary budget: 3,072 tokens/turn, 5,120 total, 2 model turns, 2 executed calls. Therefore this run is valid for activation, not for solver productivity or quality.

Arm	Tool engagement	1st-turn tool call	Median think tok/turn	Interpretation
ACT-XHIGH frozen v1.7	0.0%	0%	12,288	Reference only; catastrophic no-action path
ACT-LOW	87.5%	62.5%	715	Pass activation
ACT-MEDIUM	12.5%	0%	2,560	Weak; retire from near-term solver testing
ACT-BUDGET3072	87.5%	87.5%	3,072	Pass activation; forced-close works as an action transition

5.1 New insight: equal engagement hid very different tool quality

ACT-LOW এবং ACT-BUDGET3072 দুটোই 7/8 task-এ tools activate করেছে, কিন্তু tool behavior একই ছিল না। LOW-এর 9টি executed tool call-এর 9টিই OK; exception, timeout, guard rejection বা malformed tool turn ছিল না। BUDGET3072-এর 9 executed call-এর 5টি OK, 3 exception, 1 timeout; অতিরিক্ত 2 guard rejection এবং 3 malformed tool turn ছিল।

Probe metric	ACT-LOW	ACT-BUDGET3072
Executed calls	9	9
Executed OK	9	5

Probe metric	ACT-LOW	ACT-BUDGET3072
Tool exceptions	0	3
Tool timeouts	0	1
Guard rejections	0	2
Malformed tool turns	0	3
Stateful reuse events	2	2
Mean generated tokens/task	3,011	4,819
Final program	0	0
All-train-exact	0	0

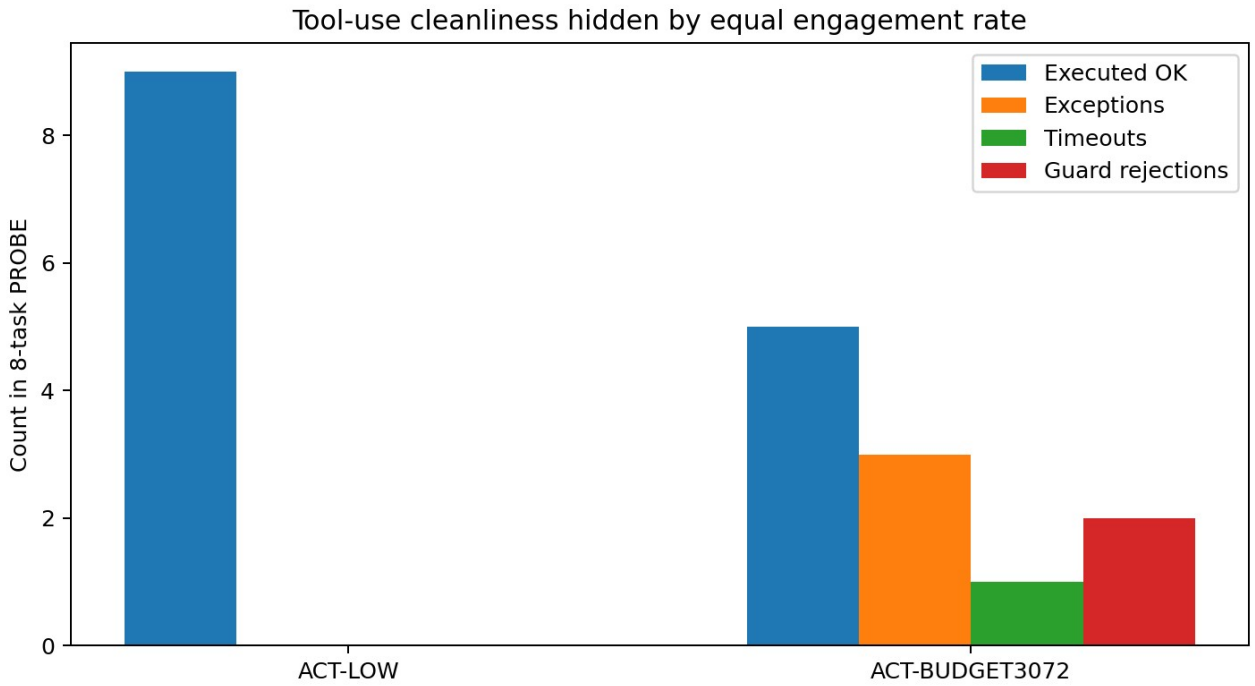


Figure 2. Equal engagement rate did not mean equal tool-use cleanliness.

INFERENCE: LOW used about 37.5% fewer generated tokens per task than BUDGET3072 in the cheap probe and produced much cleaner executed tool calls. This makes LOW the cleanest first isolation arm for C1. It does NOT prove LOW will solve more ARC tasks under the full solver budget.

6. Throughput v1.2.4 real 4xL4 audit

Run 20260916T091344Z lasted 15,452.7 seconds, about 4 h 17 m 33 s. Frozen C0 anchor reproduced the prior long-decode behavior, session drift sentinel passed, and GPU state was not marked compromised. The strongest completed engine row was C1 graph/noAR.

Config	Fixed@16 tok/s	Mixed@16 tok/s	Long@8 tok/s	Audit reading
C0 frozen Q1	37.68	23.59	48.30	enforce_eager=True; baseline

Config	Fixed@16 tok/s	Mixed@16 tok/s	Long@8 tok/s	Audit reading
I0 eager + stats	37.73	23.71	-	Instrumentation control
C1 graph/noAR	48.95	31.45	129.17	Prime solver-smoke candidate
C2 graph/custom AR	48.93	31.44	129.04	No meaningful gain over C1

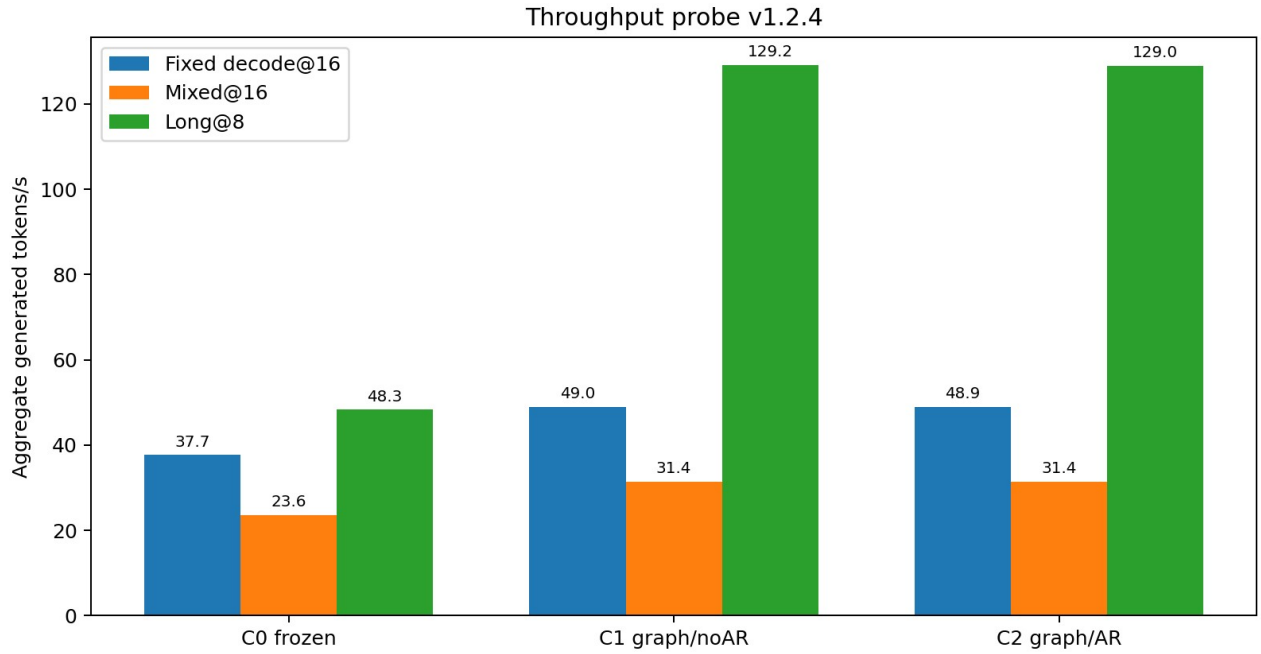


Figure 3. Real GPU throughput. C1 graph/noAR is materially faster, especially on solver-representative long decode.

C1 long@8 speedup vs frozen C0 = 129.17 / 48.30 = 2.67x. Against the instrumentation-matched eager control I0, fixed decode@16 improved about 29.8% and mixed@16 about 32.6%. Custom all-reduce changed throughput by effectively 0% at TP4, so it should remain disabled.

6.1 The automated numerics gate is invalid and must be retired

The v1.2.4 notebook still used mean greedy-prefix agreement >= 0.50 as a promotion filter. C1 scored 0.1836 and was filtered, causing the automated throughput_verdict to name C0 as the safe TP4 solver candidate. But the unchanged end-of-session C0 repeat scored only 0.1875 against the original C0 fingerprint. Therefore the fingerprint is not stable enough to establish model/solver equivalence in this environment.

RETIRED: "C0 is the solver winner because C1 fails the 0.50 prefix-agreement gate" is invalid. Raw throughput measurements remain valid; solver safety must be established by real all-train-exact / official-exact behavior, not greedy token-prefix similarity.

6.2 Prefix64 and TP2 failures: what they do and do not mean

C5 prefix64, T1 TP2 noAR এবং T2 TP2 AR all failed under aggressive capacity/concurrency settings. Logs show CUDA OOM / EngineDeadError; T2 also produced NCCL watchdog symptoms after failure. These rows test stressed capacity, not feature feasibility. TP2 reported only about 2.17x to

2.27x max concurrency at the chosen model length, yet the ladder attempted much larger concurrency points.

Row	Status	Parent wall	Interpretation
C5 prefix64	timeout	75.0 min	CUDA OOM under max_num_seqs=64
T1 TP2 noAR	timeout	75.0 min	CUDA OOM; single TP2 capacity stressed
T2 TP2 AR	failed	25.1 min	CUDA OOM + later NCCL watchdog symptoms

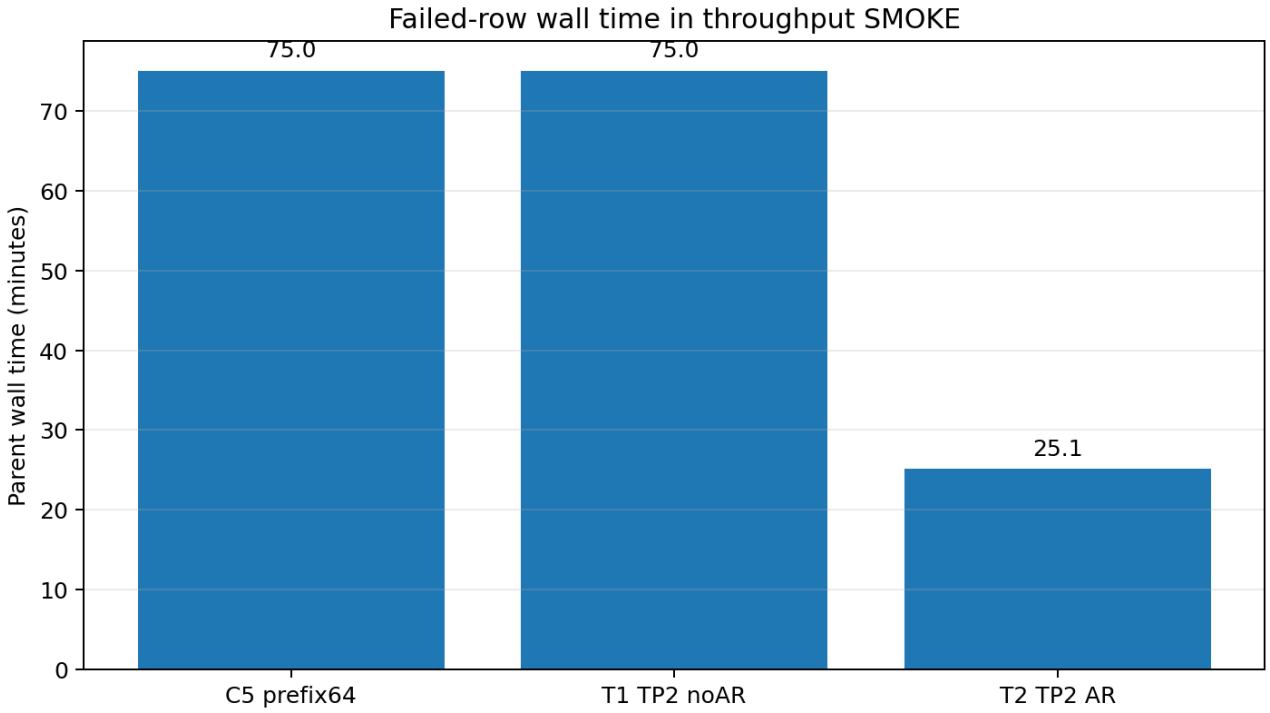


Figure 4. Failed-row parent wall time. Terminal failures were not reclaimed fast enough.

These three failed rows consumed about 175.1 minutes, approximately 68.0% of the entire 4.29 h throughput run. This confirms a parent-side fail-fast design defect: terminal worker failure was already visible, but the parent often waited for process timeout/exit instead of immediately killing and reclaiming the row.

7. P1 SPARSE side branch: useful evidence, not a main-line promotion

Two independent evidence bundles are integrity-clean and produce similar aggregate means, but the game-level details vary substantially. This branch uses a different model/agent setting and must not be mixed with the Qwen3.8-27B ARC-AGI-2 solver decision logic.

Run	Mean score	Levels	Nonzero games	Generated tokens	Actions	Runtime signal
Run A	6.9111	35	20/25	1,979,584	3,882	25 read timeouts
Run B	6.8091	39	23/25	1,934,211	3,365	24 read timeouts

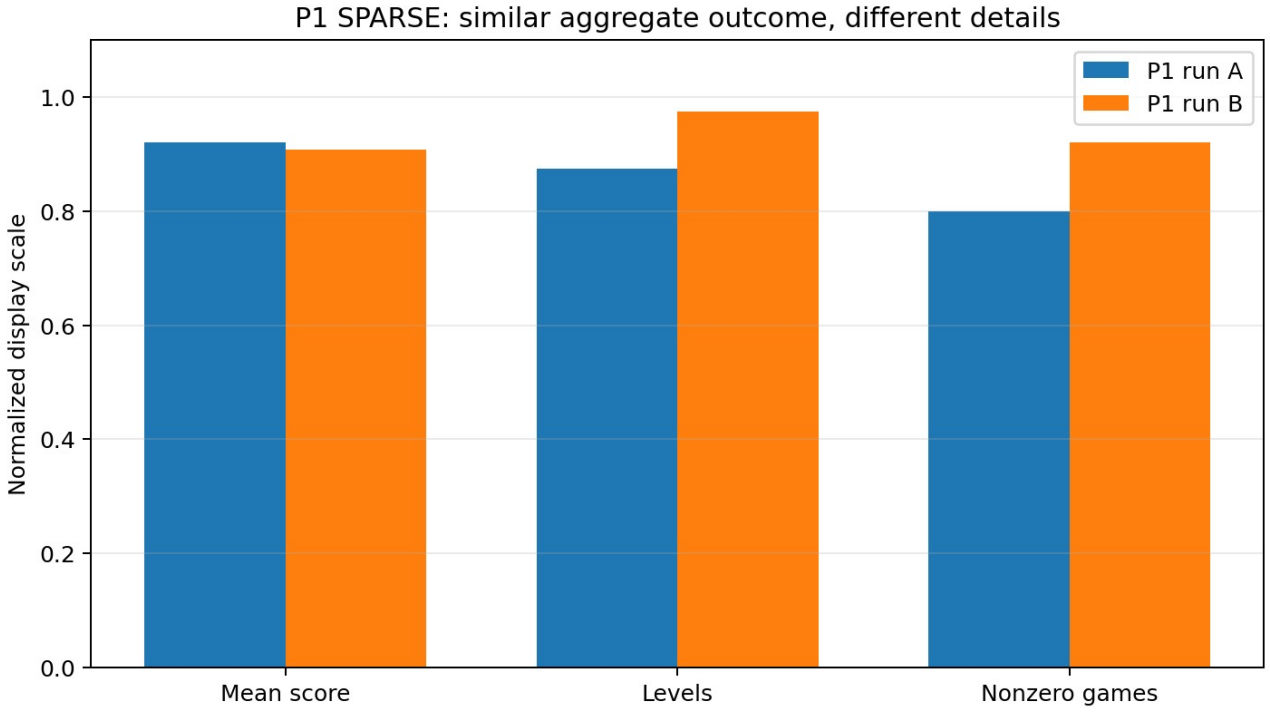


Figure 5. P1 aggregate outcome is similar, but aggregate similarity hides game-level variance.

Across the same 25 games, Run B vs Run A had 11 improvements, 9 regressions and 5 ties; per-game Pearson correlation was about 0.615. Large swaps occurred, for example ft09 dropped about 28.7 points while lp85 improved 25 points. Both runs ended all 25 game states as gave_up, and teardown evidence retained a VLLM::Worker GPU row after the server port had closed.

Decision: Keep P1 SPARSE as PROMISING / HOLD. A matched B0 vs P1 replication with controlled seeds/pairs is required before promotion. Do not import its score logic into the main Qwen ARC-AGI-2 solver branch.

8. Mistakes, invalid assumptions, and the corrected rule

Old assumption	Audit verdict	Rule that must persist
"PREFLIGHT pass = model will use tools"	False	Separate harness correctness from real model behavior. Add cheap real mechanism probes.
"Qwen reasoning_effort=high"	Wrong model contract	Use xhigh / medium / low only. Template-level assert should reject high.
Compile/nbformat pass is enough	False	Sequential/static-name checks and executed-path tests are required; v1.8 had a runtime naming bug.
Insertion-order recommendation	Bug	Rank candidates explicitly and select from ranked passing arms.
Forced-close guarantees a tool call	False	It only exits thinking; post-close output can be malformed or text.
Tool engagement = solving success	False	Promotion must follow final program -> train exact -> valid test -> official exact.
Greedy prefix >=0.50 proves safety	Invalidated	Unchanged C0 repeat also failed. Use

Old assumption	Audit verdict	Rule that must persist
		solver/exact quality gates.
Prefix/TP2 OOM = feature unsupported	False	Retest at safe concurrency/capacity before feasibility claims.
Global comparison_valid=True	Too broad	Validity should be family-level: TP4 core can be valid while prefix/TP2 families fail.
Wait for row timeout after worker terminal failure	Wasteful	Parent must monitor terminal result and fail-fast kill/reclaim.
CPU fake-engine throughput predicts GPU	False	SELFTEST validates orchestration only; real performance requires 4xL4 evidence.
Side-branch score can drive main solver	False	Keep branch metrics and causal decisions separate.

9. Canonical state after this deep audit

9.1 VERIFIED

- Q1/PIV XHIGH is a catastrophic termination pattern under the tested 12,288-token budget: v1.7 tool engagement 0/8 and all 8 die at the cap.
- LOW এবং BUDGET3072 real model tool activation 7/8 করে; MEDIUM 1/8.
- LOW probe tool calls were materially cleaner than BUDGET3072, although neither produced a final program under the cheap canary budget.
- C1 enforce_eager=False + custom AR off materially accelerates 4xL4 serving; long@8 speedup is about 2.67x vs frozen C0.
- Custom all-reduce gives essentially no throughput benefit at tested TP4 conditions.
- Current 0.50 greedy-prefix gate is invalidated by the unchanged repeat control.
- Prefix64/TP2 OOM rows are capacity-stress failures and do not establish feature infeasibility.
- Current v1.9 source notebook is code-equivalent to the executed v1.9d GPU probe except the one-line RUN_PROFILE switch.

9.2 UNKNOWN / requires experiment

- Does C1 graph/noAR preserve or improve actual ARC executable-program quality?
- Under full solver budgets, does LOW or BUDGET3072 yield higher final-program and all-train-exact rates?
- Can a healthy single lineage produce reproducible official exact solves before multi-sample scaling?
- Can prefix caching at max_num_seqs=16 improve real multi-turn wall time without OOM?
- Can TP2 fit safely at n=2/4/8 with lower scheduler/batched-token budgets, and is TP2x2 worthwhile only after that?
- What K and candidate-selection strategy maximizes verified coverage once single-lineage quality is non-zero?

10. Next decisive experiment: C1 Real Solver SMOKE

Primary question: C1 fast serving + ACT-LOW কি executable final program তৈরি করে, fresh verifier-এ all-train-exact > 0 অর্জন করে, এবং valid test output/official exact solve দিতে পারে?

10.1 Donor and causal intervention

Donor: PIV-Q27-ACT-v1.9-Reasoning-Activation-Ablation(1).ipynb, SHA256 007c8d68559c39a948dfef9d571fbd2e5da7335492a6f3330645db892648b525. This is the v1.9d source aligned with the successful PROBE_GPU code path.

```
# Frozen donor engine keys remain unchanged except this one line:
"enforce_eager": False,

# Must remain:
"disable_custom_all_reduce": True,
"enable_prefix_caching": False,
"tensor_parallel_size": 4,
"max_num_seqs": 16,
"max_num_batched_tokens": 32768,
"max_model_len": 29696,
"gpu_memory_utilization": 0.90,
"seed": 42,
```

Do not copy disable_log_stats=False from the throughput instrumentation row into the solver derivative unless it is already part of the donor. That would create a second engine intervention and weaken causal interpretation.

10.2 What must stay frozen

- Same stratified SMOKE8 task IDs and order.
- Same model artifact and runtime stack.
- Same source-aligned prompt family and Qwen tool chat template.
- Same persistent REPL, sandbox rules, retry semantics, parser and fresh verifier.
- Same seed policy and phase-specific seed logic.
- Same full solver budget for the chosen arm, not the cheap PROBE budget.
- Prefix caching OFF and custom AR OFF.
- No prompt/search/PTQ/NVARC changes in this experiment.

10.3 First arm and metrics

First isolation arm: ACT-LOW. Rationale: same 87.5% engagement as BUDGET3072 in the probe, cleaner 9/9 executed calls, zero malformed/guard/timeout, and lower token use. This isolates serving behavior with the simpler natural-low pathway.

Metric	Decision use
final_program_rate	Did the model actually emit a self-contained solution(grid)?
syntax/execution-valid rate	Can the candidate execute in a fresh process?
all_train_exact_rate	Primary quality milestone; first required non-zero signal.
test_execution_ok / valid-grid rate	Does a verified program safely produce test outputs?

Metric	Decision use
official exact	Final authority on solved outputs.
tokens/task + wall-clock/task	Required for scaling projections.
tool cleanliness + retry/timeout counts	Diagnose whether C1 changes action behavior.

11. C1 pre-run regression firewall

[ ] 01. Notebook JSON + all code cells compile.

[ ] 02. Current source differs from v1.9d donor only by explicitly documented C1 edits and run-profile selection.

[ ] 03. No reasoning_effort="high" anywhere; supported contract assert remains active.

[ ] 04. STOP_THINKING_TEXT exactness and token-exact phase-2 continuation tests remain active.

[ ] 05. Fake lineage G still asserts stateful reuse and fake traces are preserved.

[ ] 06. stop_without_think_close reaches terminal records.

[ ] 07. Guard-rejected snippets do not consume executed tool-call budget.

[ ] 08. Multi-call budget test remains dynamic after guard rejection.

[ ] 09. Train-perfect state remains separate from test execution state.

[ ] 10. Missing/duplicate terminal task IDs fail the coverage gate.

[ ] 11. No stale checkpoint/work-root contamination; run provenance and engine fingerprint written.

[ ] 12. Engine config shows enforce_eager=False, custom AR=True? NO: must be disable_custom_all_reduce=True, prefix cache=False.

[ ] 13. No throughput greedy-prefix/raw-cell similarity gate is used for correctness or promotion.

[ ] 14. PREFLIGHT success is reported separately from real GPU model behavior.

[ ] 15. Failure/terminal state should be written early enough that later harnesses can fail-fast rather than wait full timeout.

12. Decision tree after C1 SMOKE

Observed C1 outcome	Interpretation	Next action
all_train_exact > 0	Single-lineage solver quality exists	Repeat for reproducibility; then same C1 engine LOW vs BUDGET3072 on verified program quality.
final_program > 0 but train_exact = 0	Program synthesis exists but hypotheses/code are wrong	Inspect candidate failure taxonomy; improve synthesis/search/prompt in a separate registered experiment.
tool engagement high, final_program = 0	Action mechanism works, finalization/search policy is bottleneck	Focus on program-production/finalization/search architecture, not serving.
tool engagement collapses only under C1	Possible engine/numerics behavior interaction	Matched C0 vs C1 LOW smoke on same tasks/seeds; do not use token-prefix fingerprint as arbiter.
C1 infra failure/OOM	Serving integration issue	Fix engine/runtime integration before any solver conclusions.

12.1 Scaling sequence only after non-zero quality

6. Reproduce the non-zero all-train-exact result on a second controlled run.
7. Compare ACT-LOW vs ACT-BUDGET3072 on final-program, all-train-exact, official exact, token cost and cleanliness.
8. Advance to Proxy/Public-48 only after a stable solver recipe exists.
9. Add K > 1 candidate sampling with fresh train verification, code/output dedup and pass@K/oracle coverage reporting.
10. Only then run Public-120 and consider NVARC/candidate-union selection.
11. Focused throughput follow-up (prefix mns16, TP2 n=2/4/8, lower batched-token budgets) remains secondary unless solver scaling is compute-bound.

13. Canonical artifact map

Role	Artifact	SHA256 prefix	Use
Handoff PDF	ARC_AGI2_Qwen38_Project_Handoff_2026-09-16(1).pdf	7803ba4f27b71748...	Historical canonical context
v1.7 PREFLIGHT exec	experiment-1-arc-agi-2-with-qwen-3-8 (10)(2).ipynb	044dbecb835c46f3...	Harness predecessor
v1.7 SMOKE exec	experiment-1-arc-agi-2-with-qwen-3-8 (11).ipynb	453ad77ba0d5230e...	Frozen XHIGH failure reference
v1.7 SMOKE bundle	PIV_Q27_P0_SMOKE_GPU_BUNDLE(1).zip	7435b795691d1542...	Canonical XHIGH evidence
v1.9 PREFLIGHT exec	experiment-1-arc-agi-2-with-qwen-3-8_V9.ipynb	365e122e2b511ad6...	Pre-v1.9d evidence that motivated final selftest fixes
v1.9d source donor	PIV-Q27-ACT-v1.9-Reasoning-Activation-Ablation(1).ipynb	007c8d68559c39a9...	Next C1 donor
v1.9d PROBE exec	experiment-1-arc-agi-2-with-qwen-3-8 (12)(4).ipynb	59cf30009100b84a...	Real activation run
v1.9d PROBE bundle	PIV_Q27_ACT_V19_PROBE_GPU_BUNDLE2.zip	d400bbb1cbcf3bc0...	Canonical activation evidence
TP v1.2.4 exec	experiment-1-arc-agi-2-with-qwen-3-8 (12)(3).ipynb	fe43890fc83fceb1...	Real serving run
TP v1.2.4 bundle	TP_Q27_PROBE_V124_SMOKE_MASTER_EVIDENCE_20260916T091344Z(3).zip	c34fd3736dd9d256...	Canonical serving evidence
P1 run A exec	notebook6e705f697b1(1).ipynb	f0b7fe53956f5459...	Secondary branch run A
P1 run A bundle	P1_SPARSE_RUN_EVIDENCE(1).zip	311939cee6395c3f...	Secondary evidence
P1 run B exec	notebook6e705f697b (1)(2).ipynb	1be4a5b26d5099cf...	Secondary branch run B
P1 run B bundle	P1_SPARSE_RUN_EVIDENCE (1)(2).zip	d8456de66c3b9281...	Secondary evidence

14. Final reference statement

Canonical next-step reference: এই audit-এর পর primary main-line action হলো v1.9d donor থেকে ACT-LOW + C1 graph/noAR Real Solver SMOKE। Major mechanisms একসাথে পরিবর্তন করা

যাবে না। First reproducible non-zero all-train-exact result-ই next phase unlock করবে.

The project has successfully moved beyond basic model loading, parser debugging, tool-activation uncertainty and blind engine tuning. The remaining work is the score-producing part: executable hypothesis/program synthesis, exact train verification, candidate diversity and public evaluation. The next experiment must answer that question directly.

Appendix A. Audit methodology notes

- All conclusions marked VERIFIED come from uploaded executed notebooks, ZIP reports/manifests/JSON/CSV/logs, or direct source diffs.
- CPU fake-engine throughput is excluded from real performance claims.
- P1 SPARSE is explicitly separated because its task/model/score semantics differ from the main ARC-AGI-2 Qwen solver line.
- The report intentionally preserves negative results. Failed experiments are treated as causal evidence, not discarded.
- No leaderboard timing is used to infer competitors' model, hardware or serving recipe.
- No claim of ARC solve capability is made until fresh all-train-exact and official exact evidence exists.